

FastSlam 1.0 e 2.0

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FastSlam 1.0 - Overview

1: **Algorithm FastSLAM 1.0**(z_t, u_t, Y_{t-1}):

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2:   for  $k = 1$  to  $M$  do           // loop over all particles
3:     retrieve  $\left\langle x_{t-1}^{[k]}, N_{t-1}^{[k]}, \left\langle \mu_{1,t-1}^{[k]}, \Sigma_{1,t-1}^{[k]}, i_1^{[k]} \right\rangle, \dots, \right.$ 
        $\left. \left\langle \mu_{N_{t-1,t-1}^{[k]}}^{[k]}, \Sigma_{N_{t-1,t-1}^{[k]}}^{[k]}, i_{N_{t-1,t-1}^{[k]}}^{[k]} \right\rangle \right\rangle$  from  $Y_{t-1}$ 

4:      $x_t^{[k]} \sim p(x_t | x_{t-1}^{[k]}, u_t)$            // sample new pose
5:     for  $j = 1$  to  $N_{t-1}^{[k]}$  do           // measurement likelihoods
6:        $\hat{z}_j = h(\mu_{j,t-1}^{[k]}, x_t^{[k]})$            // measurement prediction
7:        $H_j = h'(\mu_{j,t-1}^{[k]}, x_t^{[k]})$            // calculate Jacobian
8:        $Q_j = H_j \Sigma_{j,t-1}^{[k]} H_j^T + Q_t$            // measurement covariance
9:        $w_j = [2\pi Q_j]^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(z_t - \hat{z}_j)^T Q_j^{-1}(z_t - \hat{z}_j)\right\}$  // likelihood of correspondence
10:    endfor

11:     $w_{1+N_{t-1}^{[k]}} = p_0$            // importance factor, new feature
12:     $w^{[k]} = \max w_j$            // max likelihood correspondence
13:     $\hat{c} = \operatorname{argmax} w_j$            // index of ML feature
14:     $N_t^{[k]} = \max\{N_{t-1}^{[k]}, \hat{c}\}$            // new number of features in map
15:    for  $j = 1$  to  $N_t^{[k]}$  do           // update Kalman filters
16:      if  $j = \hat{c} + 1 + N_{t-1}^{[k]}$  then           // is new feature?
17:         $\mu_{j,t}^{[k]} = h^{-1}(z_t, x_t^{[k]})$            // initialize mean
18:         $H_j = h'(\mu_{j,t}^{[k]}, x_t^{[k]})$ ;  $\Sigma_{j,t}^{[k]} = (H_j^{-1})^T Q_t H_j^{-1}$  // initialize covar.
19:         $i_{j,t}^{[k]} = 1$            // initialize counter
20:      else if  $j = \hat{c} \leq N_{t-1}^{[k]}$  then           // is observed feature?
21:         $K = \Sigma_{j,t-1}^{[k]} H_j^T Q_t^{-1}$            // calculate Kalman gain
22:         $\mu_{j,t}^{[k]} = \mu_{j,t-1}^{[k]} + K(z_t - \hat{z}_j)$            // update mean
23:         $\Sigma_{j,t}^{[k]} = (I - K H_j) \Sigma_{j,t-1}^{[k]}$            // update covariance
24:         $i_{j,t}^{[k]} = i_{j,t-1}^{[k]} + 1$            // increment counter

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25:   else           // all other features
26:      $\mu_{j,t}^{[k]} = \mu_{j,t-1}^{[k]}$            // copy old mean
27:      $\Sigma_{j,t}^{[k]} = \Sigma_{j,t-1}^{[k]}$            // copy old covariance
28:     if  $\mu_{j,t-1}^{[k]}$  outside perceptual
       range of  $x_t^{[k]}$  then           // should feature have been seen?
29:        $i_{j,t}^{[k]} = i_{j,t-1}^{[k]}$            // no, do not change
30:     else
31:        $i_{j,t}^{[k]} = i_{j,t-1}^{[k]} - 1$            // yes, decrement counter
32:       if  $i_{j,t-1}^{[k]} < 0$  then
33:         discard feature  $j$            // discard dubious features
34:       endif
35:     endif
36:   endif
37: endfor

38:   add  $\left\langle x_t^{[k]}, N_t^{[k]}, \left\langle \mu_{1,t}^{[k]}, \Sigma_{1,t}^{[k]}, i_1^{[k]} \right\rangle, \dots, \left\langle \mu_{N_t^{[k]},t}^{[k]}, \Sigma_{N_t^{[k]},t}^{[k]}, i_{N_t^{[k]},t}^{[k]} \right\rangle \right\rangle$  to  $Y_{\text{aux}}$ 
39: endfor

40:  $Y_t = \emptyset$            // construct new particle set

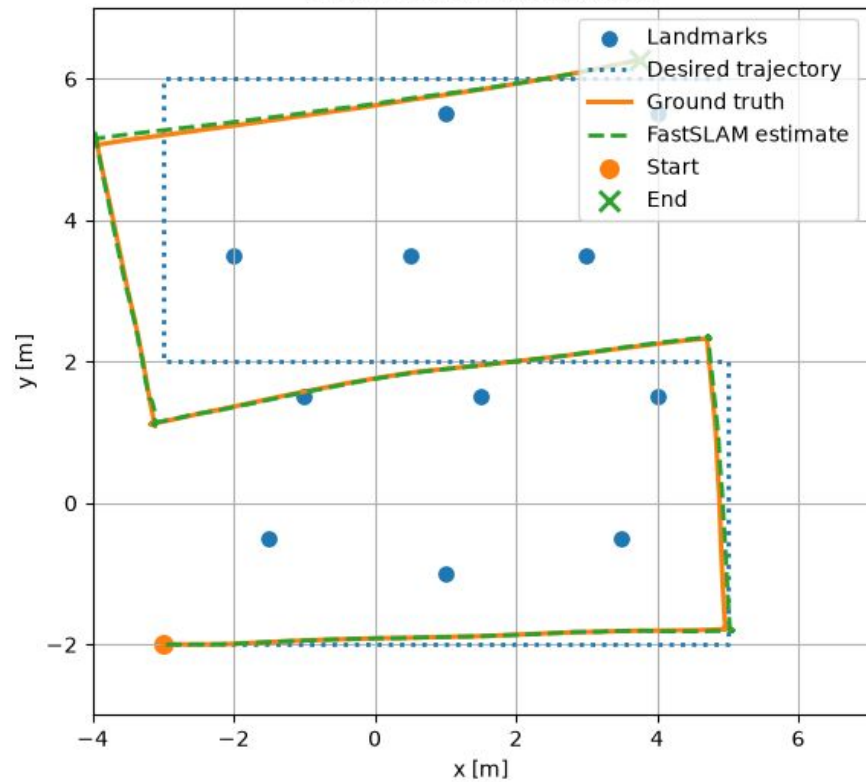
41: do  $M$  times           // resample  $M$  particles
42:   draw random index  $k$ 
   with probability  $\propto w^{[k]}$            // resample
43:   add  $\left\langle x_t^{[k]}, N_t^{[k]}, \left\langle \mu_{1,t}^{[k]}, \Sigma_{1,t}^{[k]}, i_1^{[k]} \right\rangle, \dots, \left\langle \mu_{N_t^{[k]},t}^{[k]}, \Sigma_{N_t^{[k]},t}^{[k]}, i_{N_t^{[k]},t}^{[k]} \right\rangle \right\rangle$  to  $Y_t$ 
44: enddo

45: return  $Y_t$ 

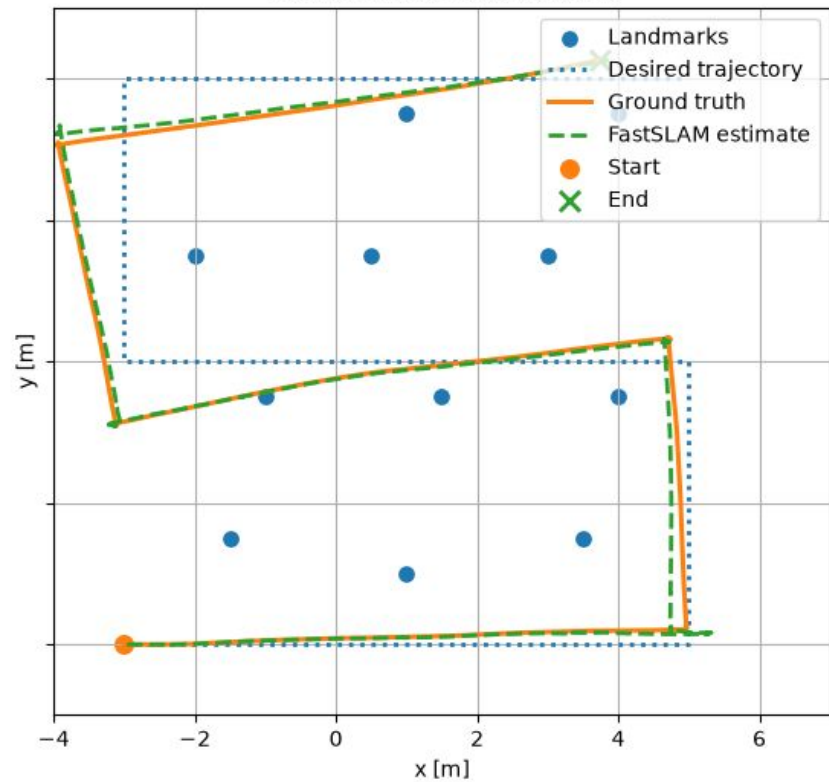
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FastSLAM Estimated Trajectory

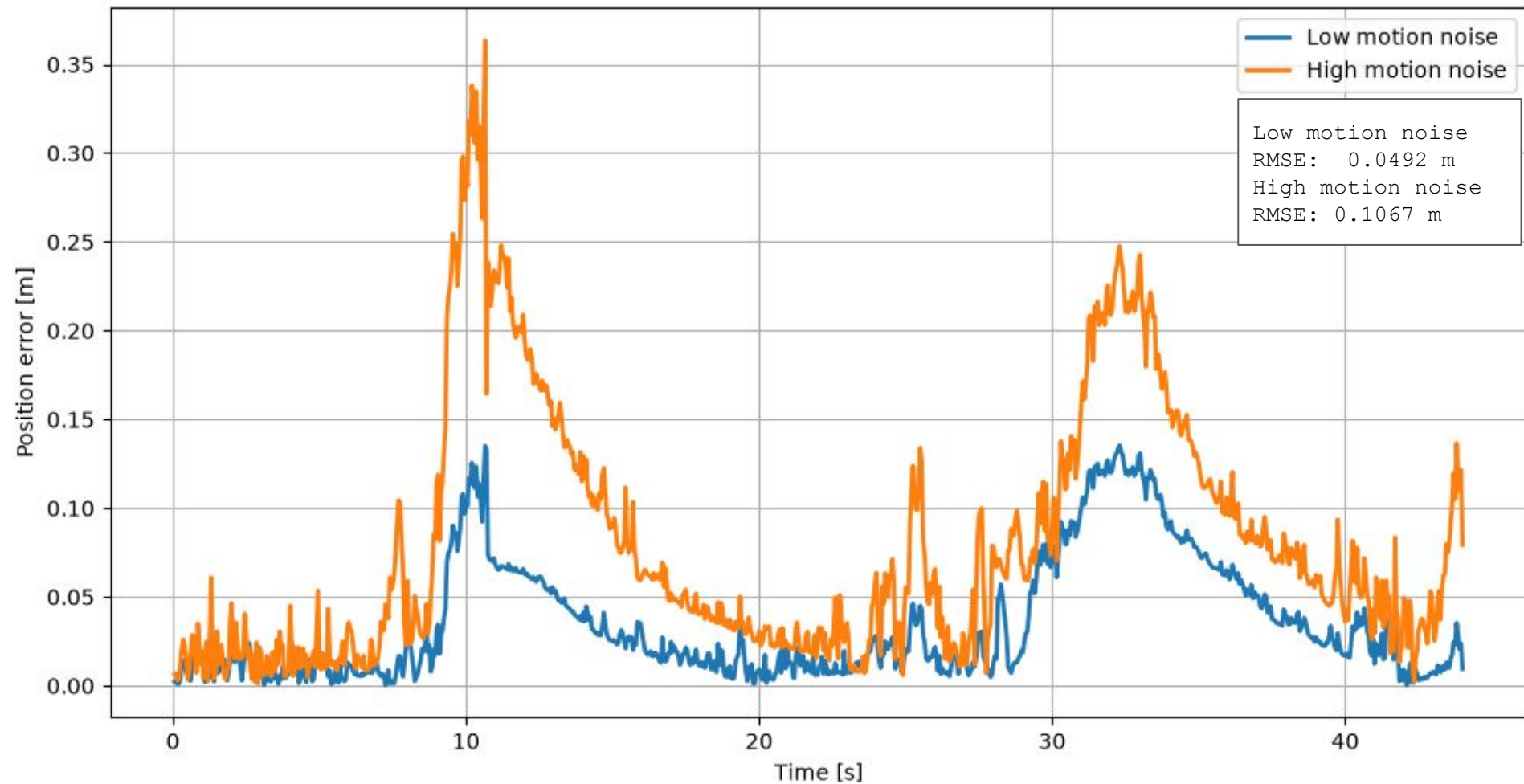
Low motion noise
Motion noise = (0.20, 0.02)



High motion noise
Motion noise = (0.50, 0.05)



FastSLAM Position Error



FastSlam 2.0 - Overview

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1: Algorithm FastSLAM 2.0( $z_t, u_t, Y_{t-1}$ ):
2:   for  $k = 1$  to  $M$  do // loop over all particles
3:     retrieve  $\left\langle x_{t-1}^{[k]}, N_{t-1}^{[k]}, \left\langle \mu_{1,t-1}^{[k]}, \Sigma_{1,t-1}^{[k]}, i_{1,t-1}^{[k]} \right\rangle, \dots, \right.$ 
        $\left. \left\langle \mu_{N_{t-1}^{[k]},t-1}^{[k]}, \Sigma_{N_{t-1}^{[k]},t-1}^{[k]}, i_{N_{t-1}^{[k]},t-1}^{[k]} \right\rangle \right\rangle$  from  $Y_{t-1}$ 
4:     for  $j = 1$  to  $N_{t-1}^{[k]}$  do // calculate sampling distribution
5:        $\hat{x}_{j,t} = g(x_{t-1}^{[k]}, u_t)$  // predict pose
6:        $\hat{z}_j = h(\mu_{j,t-1}^{[k]}, \hat{x}_{j,t})$  // predict measurement
7:        $H_{x,j} = \nabla_{x_t} h(\mu_{j,t-1}^{[k]}, \hat{x}_{j,t})$  // Jacobian wrt pose
8:        $H_{m,j} = \nabla_{m_j} h(\mu_{j,t-1}^{[k]}, \hat{x}_{j,t})$  // Jacobian wrt map feature
9:        $Q_j = Q_t + H_{m,j} \Sigma_{j,t-1}^{[k]} H_{m,j}^T$  // measurement information
10:       $\Sigma_{x,j} = [H_{x,j}^T Q_j^{-1} H_{x,j} + R_t^{-1}]^{-1}$  // Cov of proposal distribution
11:       $\mu_{x,j} = \Sigma_{x,j} H_{x,j}^T Q_j^{-1} (z_t - \hat{z}_j)$  // mean of proposal distribution
12:       $x_{t,j}^{[k]} \sim \mathcal{N}(\mu_{x,j}, \Sigma_{x,j})$  // sample pose
13:       $\hat{z}_j = h(\mu_{j,t-1}^{[k]}, x_{t,j}^{[k]})$  // measurement prediction
14:       $\pi_j = |2\pi Q_j|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (z_t - \hat{z}_j)^T Q_j^{-1} (z_t - \hat{z}_j) \right\}$  // correspondence likelihood
15:    endfor
16:     $\pi_{1+N_t^{[k]}} = p_0$  // likelihood of new feature
17:     $\hat{c} = \text{argmax } \pi_j$  // ML correspondence
18:     $N_t^{[k]} = \max\{N_{t-1}^{[k]}, \hat{c}\}$  // new number of features
19:    for  $j = 1$  to  $N_t^{[k]}$  do // update Kalman filters
20:      if  $j = \hat{c} = 1 + N_{t-1}^{[k]}$  then // is new feature?
21:         $x_t^{[k]} \sim p(x_t | x_{t-1}^{[k]}, u_t)$  // sample pose
22:         $\mu_{j,t}^{[k]} = h^{-1}(z_t, x_t^{[k]})$  // initialize mean
23:         $H_{m,j} = \nabla_{m_j} h(\mu_{j,t}^{[k]}, x_t^{[k]})$  // Jacobian wrt map feature
24:         $\Sigma_{j,t}^{[k]} = (H_{m,j}^{-1})^T Q_t H_{m,j}^{-1}$  // initialize covariance
25:         $i_{j,t}^{[k]} = 1$  // initialize counter
26:         $w^{[k]} = p_0$  // importance weight

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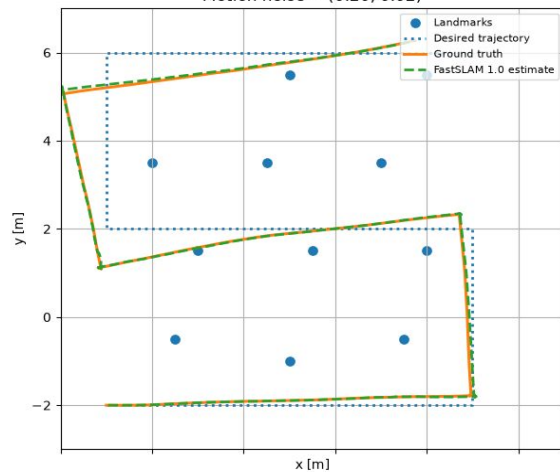
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26:       $w^{[k]} = p_0$  // importance weight
27:      else if  $j = \hat{c} \leq N_{t-1}^{[k]}$  then // is observed feature?
28:         $x_t^{[k]} = x_{t,j}^{[k]}$ 
29:         $K = \Sigma_{j,t-1}^{[k]} H_{m,j}^T Q_j^{-1}$  // calculate Kalman gain
30:         $\mu_{j,t}^{[k]} = \mu_{j,t-1}^{[k]} + K(z_t - \hat{z}_j)$  // update mean
31:         $\Sigma_{j,t}^{[k]} = (I - K H_{m,j}) \Sigma_{j,t-1}^{[k]}$  // update covariance
32:         $i_{j,t}^{[k]} = i_{j,t-1}^{[k]} + 1$  // increment counter
33:         $L = H_{x,j} R_t H_{x,j}^T + H_{m,j} \Sigma_{j,t-1}^{[k]} H_{m,j}^T + Q_t$ 
34:         $w^{[k]} = |2\pi L|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (z_t - \hat{z}_j)^T L^{-1} (z_t - \hat{z}_j) \right\}$  // importance weight
35:      else // all other features
36:         $\mu_{j,t}^{[k]} = \mu_{j,t-1}^{[k]}$  // copy old mean
37:         $\Sigma_{j,t}^{[k]} = \Sigma_{j,t-1}^{[k]}$  // copy old covariance
38:        if  $\mu_{j,t-1}^{[k]}$  outside perceptual
           range of  $x_t^{[k]}$  then // should feature have been seen?
39:           $i_{j,t}^{[k]} = i_{j,t-1}^{[k]}$  // no, do not change
40:        else
41:           $i_{j,t}^{[k]} = i_{j,t-1}^{[k]} - 1$  // yes, decrement counter
42:          if  $i_{j,t-1}^{[k]} < 0$  then
43:            discard feature  $j$  // discard dubious features
44:          endif
45:        endif
46:      endif
47:    endfor
48:    add  $\left\langle x_t^{[k]}, N_t^{[k]}, \left\langle \mu_{1,t}^{[k]}, \Sigma_{1,t}^{[k]}, i_{1,t}^{[k]} \right\rangle, \dots, \left\langle \mu_{N_t^{[k]},t}^{[k]}, \Sigma_{N_t^{[k]},t}^{[k]}, i_{N_t^{[k]},t}^{[k]} \right\rangle \right\rangle$  to  $Y_{\text{aux}}$ 
49:  endfor
50:   $Y_t = \emptyset$  // construct new particle set
51:  do  $M$  times // resample  $M$  particles
52:    draw random index  $k$ 
    with probability  $\propto w^{[k]}$  // resample
53:    add  $\left\langle x_t^{[k]}, N_t^{[k]}, \left\langle \mu_{1,t}^{[k]}, \Sigma_{1,t}^{[k]}, i_{1,t}^{[k]} \right\rangle, \dots, \left\langle \mu_{N_t^{[k]},t}^{[k]}, \Sigma_{N_t^{[k]},t}^{[k]}, i_{N_t^{[k]},t}^{[k]} \right\rangle \right\rangle$  to  $Y_t$ 
54:  enddo
55:  return  $Y_t$ 

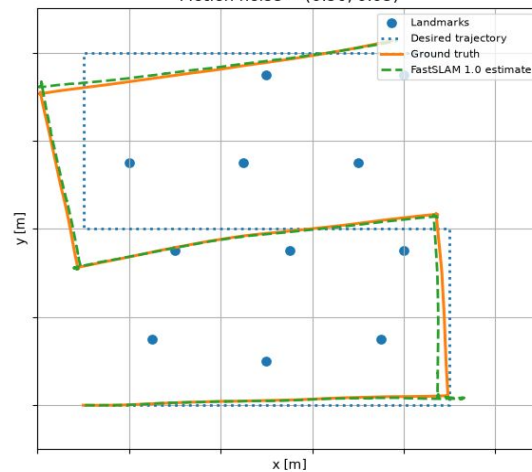
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FastSLAM 1.0 vs FastSLAM 2.0 — Estimated Trajectory

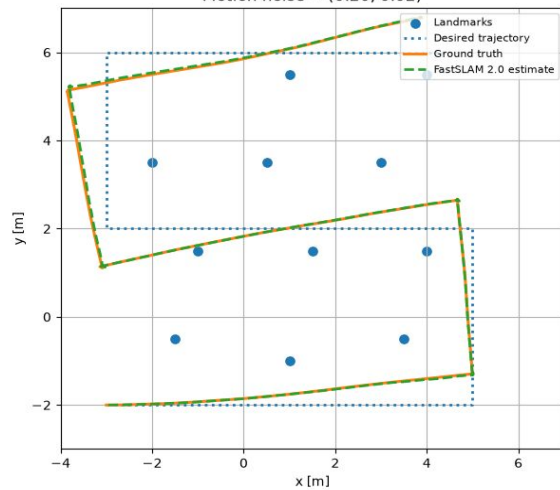
FastSLAM 1.0 — Low motion noise
Motion noise = (0.20, 0.02)



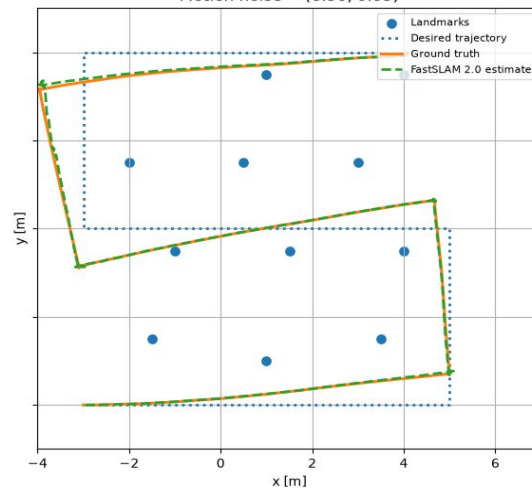
FastSLAM 1.0 — High motion noise
Motion noise = (0.50, 0.05)



FastSLAM 2.0 — Low motion noise
Motion noise = (0.20, 0.02)

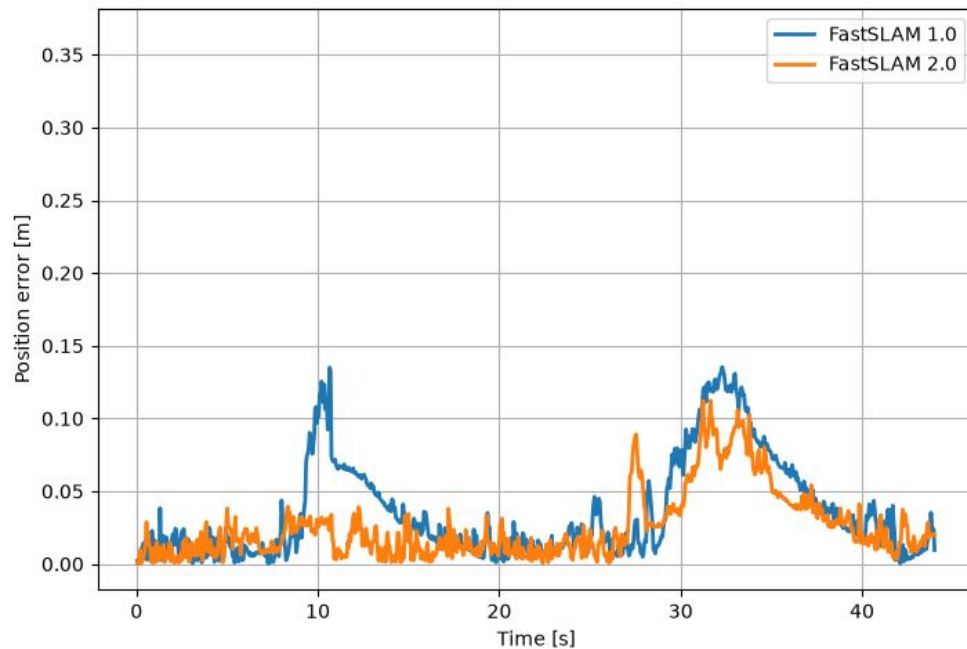


FastSLAM 2.0 — High motion noise
Motion noise = (0.50, 0.05)

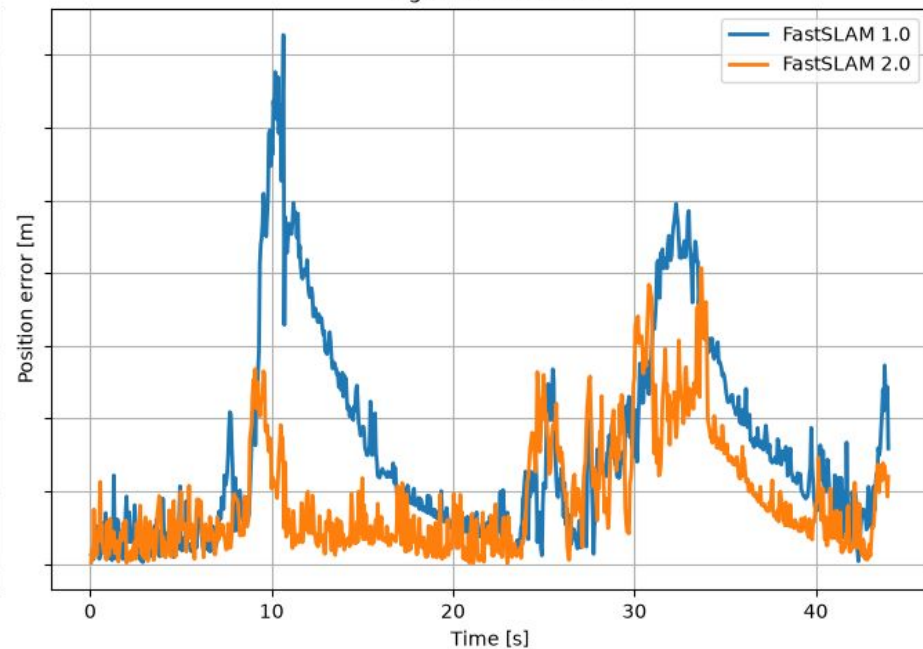


FastSLAM 1.0 vs FastSLAM 2.0 — Position Error

Low motion noise



High motion noise



Algorithm	Noise	Mean err. [m]	Max err. [m]	RMSE [m]	Median (N_{eff})
FastSLAM 1.0	Low	0.0357	0.1353	0.0492	46.7
FastSLAM 1.0	High	0.0803	0.3636	0.1067	39.9
FastSLAM 2.0	Low	0.0255	0.1123	0.0346	47.7
FastSLAM 2.0	High	0.0451	0.2033	0.0604	44.7

Conclusion

- The advantage of using fastslam 2.0 over fastslam 1.0 lies in the ability to better estimate the pose of the robot in a setup where the odometry cannot be equally trusted as the measurement.